

HQ-Net: A heatmap-based query backbone for point cloud understanding

Jun Li, Shangwei Guo, Luhan Wang, Shaokun Han*

Beijing Key Lab for Precision Optoelectronic Measurement Instrument and Technology, Beijing Institute of Technology, Beijing, China

ARTICLE INFO

Communicated by Y. Liu

Keywords:

Point cloud understanding
Deep learning
Segmentation
Classification

ABSTRACT

With the advancement of sensor technology, including LiDAR and RGB-D cameras, tasks related to point cloud understanding have gained widespread application. The sampling and learning of feature points play a crucial role in point cloud understanding, representing a fundamental and essential aspect of 3D computer vision. In the realm of point cloud processing, traditional methods such as Farthest Point Sampling (FPS) are frequently employed to select feature points. These points are then used in conjunction with effective feature aggregation operators to facilitate reasoning. However, these traditional sampling methods primarily focus on the Euclidean distance between points, lacking a comprehensive consideration of point cloud features. To address this limitation, we introduce HQ-Net, a novel point cloud understanding model designed to optimize feature point sampling using a Heatmap-based Query Backbone. The Heatmap-based Query Backbone utilizes the Heatmap Module to select category-specific point features, effectively generating a heatmap tailored to the category features and automatically extracting features from the heatmap. Subsequently, the proposed Query Encoder is employed to facilitate the interaction between point features and query features, providing mutual guidance and enhancing the model's ability to capture spatial structure and semantic information within point clouds. Additionally, we propose task-specific heads, such as classification and segmentation heads for different tasks, which can be seamlessly integrated with the backbone to achieve excellent performance. In various point cloud understanding tasks, including shape classification, part segmentation, and semantic segmentation, HQ-Net demonstrates excellent performance.

1. Introduction

With the advancement of LiDAR, RGB-D, and other sensors, the efficiency of three-dimensional data acquisition has significantly improved. This development sparks increasing interest in point cloud data and its associated understanding tasks, which find extensive applications across various domains, including autonomous driving [1], semantic segmentation [2], augmented reality [3], and consumer and medical fields [4]. For instance, in the field of autonomous driving, it is crucial to accurately detect vehicles [1] and identify pedestrians [5,6]. Deep learning methods enable the direct recognition of different objects from point clouds obtained via scans from vehicle-mounted LiDAR systems [7–11]. Flexible actuators are popular in consumer and medical fields; however, they are often difficult to model due to their viscoelasticity and nonlinearity. Using point cloud deep learning techniques to calibrate simulations of soft robots so that they are consistent with the actual deformations of real robots is a promising approach [4,12–14]. Robots modeled using point clouds, such as controlling creeping robots inspired by inchworms [15], have played a role in the transportation field.

Facing challenges associated with 3D point cloud data, such as sparsity, irregularity, and disorder, researchers explore various methods to address three-dimensional point cloud understanding tasks. These methods include multi-view networks [16,17], voxel-based networks [18–21], and point-based networks [11,22–25]. Multi-view networks project point clouds from multiple viewpoints into regular formats for 2D convolution operations. Similarly, voxel-based networks convert point clouds into 3D voxels, followed by 3D convolutions for feature extraction. However, voxel-based networks face computational costs related to voxel processing [26]. To address this, some voxel-based networks [18,27] employ sparse convolution to enhance computational efficiency by performing convolutions only on sparse data, optimizing resource usage and improving voxel processing efficiency [27]. In point-based networks, PointNet [28] directly processes unordered point cloud data using Multi-Layer Perceptron (MLP) layers and symmetric functions (e.g., max pooling and average pooling) for global feature extraction. However, PointNet [28] struggles to capture local geometric information from the spatial relationships of point clouds. To address this issue, PointNet++ [29] introduces convolution on grouped neighboring points to learn local features and enhance the

* Corresponding author.

E-mail addresses: junl@bit.edu.cn (J. Li), 3120195346@bit.edu.cn (S. Guo), 3120210613@bit.edu.cn (L. Wang), skhan@bit.edu.cn (S. Han).

<https://doi.org/10.1016/j.neucom.2024.128413>

Received 20 November 2023; Received in revised form 8 August 2024; Accepted 14 August 2024

Available online 17 August 2024

0925-2312/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

performance of point cloud understanding tasks. In recent years, transformers [11,25,30,31] have shown outstanding performance in point cloud understanding tasks, capturing relationships between any pair of features regardless of their positions in Euclidean space. Transformers can effectively capture long-distance features. An example is the Point Cloud Transformer (PCT) [25], which uses offset attention mechanisms to learn global features from point clouds.

Among these methods, some point cloud understanding models [7, 23,29,32] adopt hierarchical structures for feature learning. They sample points from point clouds layer by layer and design neural network-based local feature aggregation operators for learning feature from point sets. Therefore, the sampling and learning of feature points play a crucial role in point cloud understanding tasks, representing a fundamental and essential task in 3D computer vision.

Methods for sampling from point clouds can be categorized into traditional methods and learning-based methods. Traditional methods include random sampling (RS), grid sampling, uniform sampling, and geometric sampling, among other mathematical and statistical sampling methods. Farthest Point Sampling (FPS) [33,34] is a common method in traditional approaches. FPS achieves sampling by selecting a group of points that are farthest from each other. However, FPS only considers the Euclidean distance between points, lacking a comprehensive consideration of point cloud features. With the development of deep learning, some learning-based sampling methods have emerged, such as SNet [35] and SampleNet [36]. These methods generate new point cloud sets of the desired size as feature points through a generative approach and, in combination with different feature learning operations, achieve significant performance improvements in point cloud understanding tasks.

In the application of point clouds, similar to CenterNet [37] converting 2D images into 2D heatmaps, CenterPoint [38] maps 3D point clouds to 2D space and utilizes 2D heatmaps for object detection. However, storing point cloud information in 2D heatmaps in 3D space may lead to the loss of spatial structure and semantic information, as 2D heatmaps cannot accurately reflect information in the vertical dimension. To comprehensively perceive the spatial structure of point clouds and extract key feature points, we introduce an innovative point cloud understanding network, **HQ-Net**. This method assigns scores to each point, generates heatmaps, automatically selects query features, and eliminates the need for manual selection or traditional downsampling methods. Subsequently, we use the designed feature aggregation operator to infer query features and combine them with specific task heads to achieve outstanding performance.

Specifically, HQ-Net consists of the Heatmap-based Query backbone and task-specific heads. Our proposed backbone comprises the Heatmap Module and stacked Query Encoders. It initially extracts point features from the point cloud. Using the Heatmap Module, it dynamically adjusts the scores of point features by learning weights and biases, thereby selecting features specific to the category and generating category-specific heatmaps. This innovative approach allows automatic selection of query features based on heatmaps, enhancing the model's performance in point cloud understanding. Subsequently, the Query Encoder facilitates mutual guidance in learning between point features and query features, improving the model's perception and understanding of point cloud details and structures. The Query Encoder includes transformer decoders with embedded Multi-scale Point Local Attention (T_D -MS-PLA) and Cross Attention (T_D -CA). T_D -MS-PLA aggregates multi-scale local features of query features from point features, effectively learning multi-scale local information of query features. This comprehensive consideration of local information aids in capturing details at different spatial scales in the point cloud. Furthermore, T_D -CA, based on the similarity matrix between point features and query features, achieves information interaction between them, updating point feature information. This process calculates the degree of correlation between point features and query features, adjusts weights based on these correlations, allowing query features to guide and influence the

update of point features. For specific point cloud understanding tasks, such as shape classification and segmentation, we design task-specific heads, including the classification head and the segmentation head, which rely on these learnable point features and query features for prediction.

HQ-Net is evaluated across various point cloud understanding tasks, including shape classification, part segmentation, and semantic segmentation. Experimental results demonstrate that the Heatmap-based Query Backbone integrates well with different task heads and achieves state-of-the-art (SOTA) performance. Our contributions can be summarized as follows:

- We propose HQ-Net, a novel network for point cloud understanding, consisting of a Heatmap-based Query Backbone and task-specific heads tailored for various tasks, achieving satisfactory performance.
- We introduce a Heatmap Module that dynamically adjusts the scores of points to select category-specific features, effectively generating heatmap suitable for the category. This adaptive query feature selection method enables HQ-Net to flexibly extract key information from point clouds without manual selection or traditional downsampling methods.
- We propose a transformer-based module, Query Encoder, allowing point features and query features to guide each other. T_D -MS-PLA aggregates multi-scale local features of query features, while T_D -CA achieves information interaction between point features and query features, improving the model's ability to capture spatial structure and semantic information in point cloud understanding tasks.
- Experimental results show that HQ-Net integrates well with different task heads, achieving state-of-the-art performance in various point cloud understanding tasks.

2. Related work

2.1. Point cloud sampling

Point cloud sampling, as a fundamental and crucial task in 3D computer vision, has garnered significant attention. Traditional point cloud sampling methods primarily rely on non-learning-based approaches, with Farthest Point Sampling (FPS) [33,34] being the most widely used. The core idea of FPS is to initiate from an initial point and progressively select the farthest point from the currently chosen points based on Euclidean distance, ensuring a uniform distribution of sampled points in space. Typically, FPS is combined with local feature aggregation operators for point cloud feature extraction. For instance, PointNet++ [29] introduces the Set Abstraction (SA) module to better capture local and global features of point clouds. SA utilizes FPS to hierarchically sample the input point set to obtain feature points, followed by Multi-Layer Perceptrons (MLP) to extract local features. FPS is computationally expensive with a complexity of $O(N^2)$. In contrast, Inverse Density Importance Sampling (IDIS) [39] sorts points based on density and selects the top K points with a lower complexity of $O(N)$. Random Sampling (RS), the simplest method, has a constant complexity of $O(1)$, independent of the number of points.

However, FPS only takes into account the Euclidean distance between points, lacking a comprehensive consideration of point cloud features. In recent years, a series of learning-based point cloud feature sampling methods have emerged, showcasing superior performance in point cloud understanding tasks. The pioneering work of SNet [35] directly generates new point coordinates from global representations. KC-Net [40] employs graph-based max-pooling for downsampling, where the max-pooling operates on a pre-constructed k-NN graph (KNNG) to obtain maximum features in each node's neighborhood. Critical Points Layer (CPL) [41] adaptively downsamples points based on learned features, filtering out unimportant points while retaining crucial ones according to their contribution levels to global max-pooling.

In this work, to address the limitations of FPS on point cloud features, we propose the Heatmap Module. This module dynamically adjusts the scores of point features by learning weights and biases, selecting class-specific features to effectively generate a heatmap applicable to those class-specific features. Through this innovative approach, we achieve a more comprehensive consideration of point cloud features and the automatic selection of query features based on the heatmap, significantly enhancing the performance of our point cloud understanding model.

2.2. Deep learning on point clouds

When working with three-dimensional point cloud data, researchers have utilized a variety of network architectures to enhance their capacity for understanding and processing this type of data. These architectures can be classified into three categories: voxel-based networks, point-based networks, and transformer-based networks.

2.2.1. Voxel-based network

Voxel-based methods [20,21], utilized for handling unordered point cloud data, initially divide the three-dimensional data into regular voxel grids and allocate each point in the point cloud to the corresponding voxel grid. Subsequently, through the application of three-dimensional convolutional neural networks (3D-CNNs), feature learning is performed on this voxelized data to extract semantic features from the point cloud.

However, voxel-based approaches also face several challenges and limitations [26]. On one hand, low-resolution voxelization may result in loss of detail. When the size of the voxels is large, small detailed structures may not be accurately captured, which impacts the model's representational ability of point cloud details. On the other hand, high-resolution voxelization brings about a tremendous computational burden as it increases the parameter count and computation in the model. Despite these challenges, voxel-based methods are still widely used in 3D object detection [7,42,43] because the voxelized representations reduce redundancy in the data, effectively handling large-scale point cloud datasets.

To alleviate the computational costs brought by 3D convolution, CNNs composed of sparse convolutions are employed to extract voxel features. OctNet [44] leverages the sparsity of input data, hierarchically segmenting the space using an unbalanced octree. A node of the octree can be represented as a voxel grid. If there is point cloud data within the voxel grid, the voxel is further divided into eight smaller sub-voxels or leaf nodes. This subdivision process continues until no more point cloud data is contained within the leaf nodes. Despite the existence of cases where OctNet [44] learns features over empty voxels, it alleviates computational load to some extent compared to dense 3D convolution. [27] introduces the submanifold sparse convolution, obviating the need for computations in empty areas. When the receptive field's center covers an active input site, it executes convolution operations by creating a hash table to store computational results in specific locations.

In summary, voxel-based methods, despite the trade-off between processing complexity and performance, hold a critical role in handling large-scale point cloud datasets.

2.2.2. Point-based network

PointNet [28] is an innovative method for handling unordered point cloud data, primarily employing multi-layer perceptron (MLP) layer to learn features of each point and using symmetric functions (such as max pooling or average pooling) to aggregate global features. However, PointNet [28] is not adept at effectively capturing the local structure and detailed information of point clouds. To address this issue, PointNet++ [29] is proposed, which is based on a hierarchical network architecture. Unlike PointNet [28], which directly aggregates information on a global scale, PointNet++ [29] employs a query ball operation. This operation creates a local region within a certain radius

centered on a point, with PointNet [28] then being applied to extract features from each local region. Such a design enables PointNet++ [29] to learn both global and local features simultaneously, thereby providing a more profound understanding of complex three-dimensional shapes.

Although the PointNet-based models have achieved significant results, researchers have further explored how to define convolution operations on point clouds to better understand them. For instance, PointConv [45] contends that point cloud convolution should be a weight function related to location and content. It consists of a convolution kernel in 3D space directly composed of a weighted function learned by MLP and a density function based on kernel density estimation. Moreover, PAConv [24] proposes a position adaptive convolution method, using a ScoreNet to generate a weight matrix for each point within a local neighborhood. This weight matrix is used to construct the convolution kernel and is adaptively adjusted to dynamically change according to the point cloud data. Compared to traditional 2D convolution, PAConv [24] builds convolution kernels through this data-driven approach, achieving greater flexibility and enhanced expressive capacity. SC-CNN [46], proposed by Spatial Covering Convolutional Neural Network (SC-CNN), implements depth-separable convolution on point clouds. The spatial covering operator (SCOP) replaces the depth convolution. By stacking SC-Conv to construct SC-CNN with a hierarchical network structure for point cloud analysis, the shape information of the point cloud can be better perceived, and the robustness of the network can be improved.

2.2.3. Transformer-based network

Initially developed for Natural Language Processing (NLP) [47–50], transformers [51,52] achieve significant success. The success [53–55] not only revolutionize the field of computer vision (CV) but also spark a growing interest in applying Transformer architectures to the processing of 3D point clouds.

Point Transformer (PT) [11] uses a self-attention mechanism to build a hierarchical structure to capture the global information of point clouds and improve the performance of tasks such as point cloud classification. Point Cloud Transformer (PCT) [25] focuses on extracting local features and learns global features through the offset attention-based transformer, which enables it to effectively capture spatial relationships and semantic information in point clouds. 3D Convolutional Transformer Network (3DCTN) [56] combines the powerful local feature learning ability of convolution and the excellent global context modeling ability of transformer. 3DCTN [56] uses Graph Convolution to aggregate multi-scale local features, uses Transformer and learns global features. 3DCTN [56] has fewer parameters and FLOPS (floating point operations) than other Transformer methods such as PT. Furthermore, PU-Transformer [57] is the first work that attempts to use Transformer for point cloud upsampling. PU-Transformer [57] focuses on capturing local feature information and establishing the relationship between the outputs of the multi-head self-attention mechanism. The Spatial Depth Attention Network (SDANet) [58] is designed to improve the performance of point cloud classification and segmentation tasks. It integrates a local depth attention mechanism within the MLP to acquire local geometric representations of neighborhood points. To optimize feature extraction, these local deep attention features are fused with global features to provide a comprehensive understanding of the point cloud data. Spatial and Channel Attention-based network (SCA) [59] leverages the PointAT and AT modules to extract meaningful geometric information. It achieves this by utilizing the geometric relationships between point cloud patches, introducing a point attention model grounded in spatial and channel attention to learn the geometric relationships among point clouds. PointGT [60] originates from the integration of local abstraction (LA) and optimization strategies. The bi-directional cross-attention mechanism merges these two components into local channels, with the main goal of optimizing the utilization of edges and internal contours.

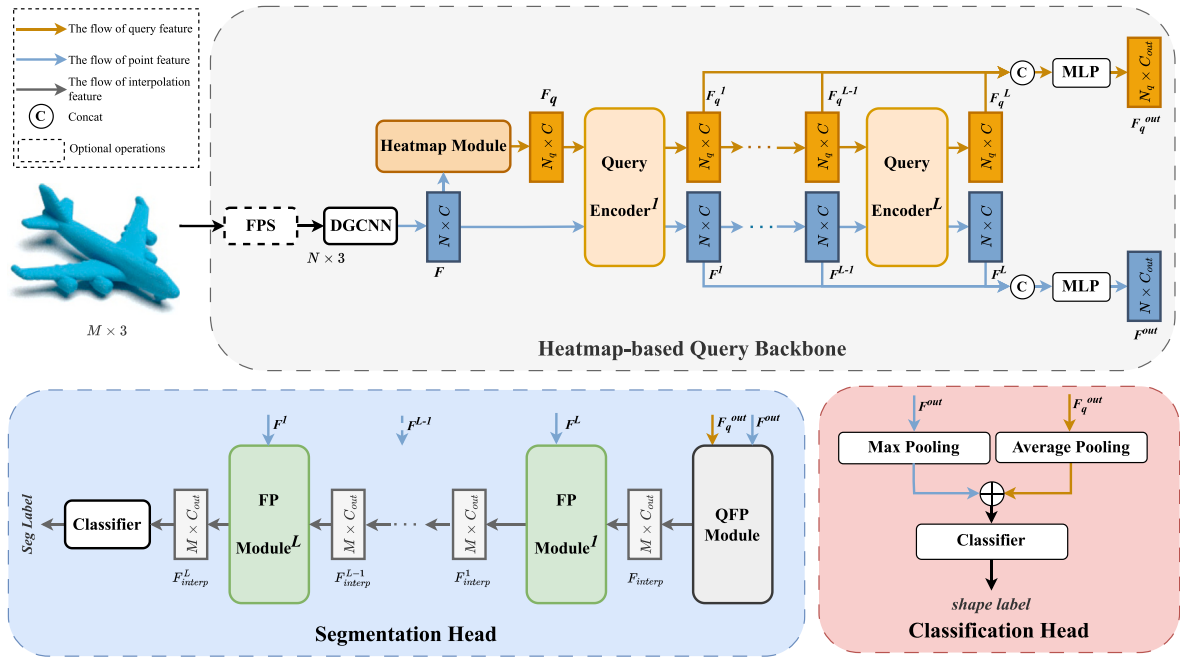


Fig. 1. The overall architecture of HQ-Net. HQ-Net employs a Heatmap Module to generate heatmaps that adaptively select query features. Meanwhile, we design a transformer-based Query Encoder to learn the interaction between the point features and query features, thereby inferring the query features. Furthermore, for different point cloud understanding tasks, we design task-specific heads, such as classification heads and segmentation heads for prediction.

3. Method

HQ-Net is a framework developed to solve the task of 3D point cloud understanding. To achieve this goal, HQ-Net is designed to contain a Heatmap-based Query Backbone as well as specialized task heads, including a classification head and a segmentation head. We introduce the proposed Heatmap-based Query Backbone in Section 3.1 and the specialized task heads in Section 3.2.

3.1. Heatmap-based query backbone

In the context of heatmap-based query backbones, the application of an optional operation, Farthest Point Sampling (FPS), marks an initial and critical step in the workflow, as shown in Fig. 1. Despite the acknowledged limitations of FPS, particularly its reliance solely on Euclidean distance, it is intentionally chosen for its simplicity and demonstrated computational efficiency in selecting a representative subset of points [29]. This strategic utilization of FPS is driven by the overarching goal of the downsampled point cloud P' , created through FPS, which aims to alleviate the computational burden while retaining salient features for subsequent analysis. For the input point cloud, we use FPS to sample a fixed number of points N . If the current number of point cloud points is less than N , then this operation is not performed. Following this downsampling step, a lightweight Dynamic Graph CNN (DGCNN) [61] is systematically applied to extract point features $F \in \mathbb{R}^{N \times C}$ from the downsampled point cloud P' . This strategic choice enhances the model's capability to capture crucial local geometric information, laying a solid foundation for the subsequent stages of the workflow. Subsequently, the introduction of the Heatmap Module becomes instrumental. The primary objective of this module is to learn and generate a category-specific heatmap based on the extracted point features F . Despite the acknowledged limitations of FPS, its initial use serves a specific purpose in simplifying the point cloud while maintaining representativeness. The Heatmap Module plays a pivotal role in dynamically adjusting the heatmap, considering the spatial relationships within the point cloud. This dynamically adjusted heatmap functions as a guide for the subsequent extraction of query features $F_q \in \mathbb{R}^{N_q \times C}$. In essence, we emphasize the strategic reasoning behind the

choice of FPS and the subsequent introduction of the Heatmap Module. The combined approach aims to balance simplicity, computational efficiency, and feature representativeness, demonstrating a thoughtful and purpose-driven strategy in addressing the challenges of 3D point cloud understanding tasks.

To further learn and extract multi-level query features and point features, we introduce L Query Encoders. For the l th Query Encoder ($l = 1, 2, \dots, L$), it utilizes the previous $F_q^{l-1} \in \mathbb{R}^{N_q \times C}$ and $F^{l-1} \in \mathbb{R}^{N \times C}$ as inputs to generate $F_q^l \in \mathbb{R}^{N_q \times C}$ and $F^l \in \mathbb{R}^{N \times C}$. Specifically, we set the initial query features F_q^0 to be F_q obtained by Heatmap Module, and the initial point features F^0 to be F obtained by DGCNN [61]. The outputs of these L Query Encoders are sequentially extracted and then concatenated at varying levels. These concatenated features are subsequently processed through the MLP, which generates final outputs $F_q^{out} \in \mathbb{R}^{N_q \times C_{out}}$ and $F^{out} \in \mathbb{R}^{N \times C_{out}}$ for the Heatmap-based Query Backbone, where $C_{out} = C \times L$.

Heatmap-based Query Backbone captures important information in point cloud through the Heatmap Module to extract query features, and then uses stacked Query Encoders to learn multi-level features. The process of multi-level features learning allows the HQ-Net to capture different levels of information in the point cloud and understand the semantic content. For the study of the relationship between Query Encoder and multi-level features, we explored it in Section 5.3.

3.1.1. Heatmap module

In the domain of feature selection, it is paramount to meticulously identify a subset of features that encapsulates the most informative attributes pertinent to the target variable. Mutual information, represented as $I(X; Y)$, quantifies the interdependence between a set of features X and class labels Y . According to the definition of entropy, $I(X; Y)$ can also be expressed as:

$$I(X; Y) = H(Y) - H(Y|X) \quad (1)$$

where, $H(Y)$ is the entropy of Y , and $H(Y|X)$ is the conditional entropy of Y given X .

For any candidate subset $X' \subseteq X$, the mutual information with respect to class label Y is computed as $I(X'; Y)$. Our objective is to

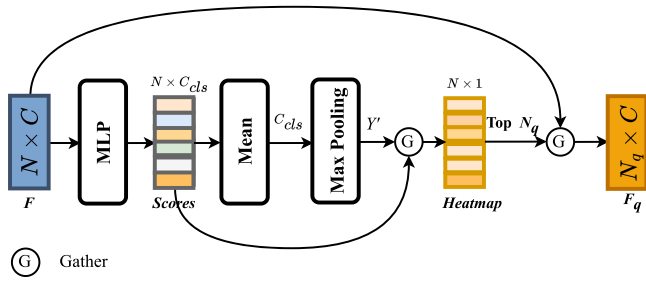


Fig. 2. Illustration of the Heatmap Module. The Heatmap Module generates a heatmap to adaptively select query features. It first predicts the class label based on the point features F , and then selects the top N_q features with the corresponding class label as the query features F_q .

identify the optimal subset $X_k \subseteq X$. The objective is to determine the optimal subset $X_k \subseteq X$, ensuring that the selected features are highly relevant to the target variable while minimizing the presence of redundant or irrelevant data. This process can be expressed by Eq. (2). This strategy ensures an effective balance between the comprehensiveness and efficiency of the feature selection process.

$$I(X_k; Y) = \max \{I(X'; Y)\} \quad (2)$$

Combined with Eq. (1), Eq. (2) can be expressed as:

$$I(X_k; Y) = H(Y) - H(Y|X_k) \quad (3)$$

This stems from the fact that $H(Y)$, the entropy of the class labels Y , is a constant value determined solely by the underlying data distribution and remains invariant across different models. In contrast, $H(Y|X_k)$, the conditional entropy of Y given the features X_k , is modifiable through the learning process of the model. By optimizing the model to reduce $H(Y|X_k)$, we enhance the model's predictive certainty regarding the output Y . Consequently, maximizing the mutual information $I(X_k; Y)$ effectively translates into an optimization problem where $H(Y|X_k)$ is minimized. This reduction in conditional entropy directly corresponds to an increase in the information shared between the features X_k and the class labels Y , thus improving the overall predictive performance of the model.

While Farthest Point Sampling (FPS) is a common sampling method, it has limitations as it solely relies on the Euclidean distance between points, disregarding feature information. To overcome this and enhance the extraction of representative points for point clouds, we have designed a sampling strategy based on deep learning. This strategy involves generating a heatmap by learning the spatial relationships between points. Subsequently, features are dynamically selected based on this heatmap. This approach is aimed at improving the representation of selected points and achieving enhanced performance compared to traditional methods like FPS.

The Heatmap Module possesses the capability to learn weights and biases based on the point features F obtained through DGCNN [61]. Subsequently, it generates a category-specific heatmap, which is then employed to dynamically select crucial features within the point cloud. The learning of weights and biases enables the model to dynamically adjust the scores of $F \in \mathbb{R}^{N \times C}$. These scores play a pivotal role in determining the numerical distribution within the generated heatmap, representing the relative importance of different elements in F . By effectively ranking the features according to their relevance to the specific categories, the heatmap guides the feature selection process. Consequently, this ranking mechanism ensures that the selected features maximize the mutual information between the feature subset and the class labels, thereby enhancing the model's predictive accuracy and efficiency.

As illustrated in Fig. 2, we generate the map $Scores \in \mathbb{R}^{N \times C_{cls}}$ by applying MLP to the point features F , where C_{cls} represents the number

of shape labels. Following the generation of $Scores$, the model applies an averaging operation to condense these into a one-dimensional vector of length C_{cls} . Each element of this vector represents the aggregated score for a specific category, summarizing the overall propensity of the data to associate with each category. The generated heatmap guides the selection of the top N_q features from F , forming the query features $F_q \in \mathbb{R}^{N_q \times C}$.

When the model utilizes the heatmap to select query features F_q , the mutual information between F_q and the predicted labels Y' reaches its maximum, denoted as $I(F_q; Y')$. This maximization corresponds to minimizing the conditional entropy $H(Y'|F_q)$, reflecting a reduction in uncertainty about the class labels given the selected features. This minimized conditional entropy aligns directly with the objective of cross-entropy loss, which aims to reduce the discrepancy between the predicted probabilities $\hat{p}(y'|x)$ and the actual data distribution $p(y|x)$.

The Heatmap Module, by learning weights and biases, dynamically adjusts the importance scores of elements in F . Furthermore, it facilitates class-specific feature selection. Based on the predicted class, specific scores are extracted from $Scores$, effectively generating a heatmap tailored to the characteristics of that class. This heatmap is then employed to select the top N_q features from the original F as query features. This meticulous selection process ensures the utilization of the most relevant features for a given category in the subsequent feature extraction. To illustrate the effectiveness of our proposed Heatmap Module, we conduct ablation experiments on different sampling methods in Section 5.1. Additionally, the relationship between the Heatmap Module and the number of query features is explored in Section 5.2.

3.1.2. Query encoder

To extract query features F_q from point features F , we employ the Heatmap Module. For effective mutually guided learning between F and F_q , we introduce a transformer-based Query Encoder. Transformers demonstrate the capability to capture global context information effectively, transcending local receptive fields. Utilizing the attention mechanism, Transformers empower the model to dynamically focus on information from various locations during the learning process.

As depicted in the (a) of Fig. 3, the l_{th} Query Encoder is comprised of two main transformer decoders, where $l = 1, 2, \dots, L$. The first transformer decoder incorporates Multi-scale Point Local Attention (MS-PLA), denoted as T_D -MS-PLA. The subsequent decoder implements cross-attention (CA), denoted as T_D -CA. T_D -MA-PLA takes input from $F^{l-1} \in \mathbb{R}^{N \times C}$ and $F_q^{l-1} \in \mathbb{R}^{N_q \times C}$. Utilizing the spatial information of the point cloud, it extracts multi-scale neighborhood features of F_q^{l-1} from F^{l-1} , capturing local geometric information and resulting in $F_{local} \in \mathbb{R}^{N_q \times C}$. Subsequently, T_D -CA facilitates the interaction between F_{local} and F^{l-1} based on the similarity matrix of point features and query features, thereby updating point feature information and obtaining $F^l \in \mathbb{R}^{N \times C}$, which can be expressed as:

$$F_{local} = T_{D-MS-PLA}(\sigma(F_q^{l-1}), F^{l-1}) \quad (4)$$

$$F^l = T_{D-CA}(F^{l-1}, F_{local}) \quad (5)$$

where, σ represents an MLP layer.

Subsequently, a Max-Pooling layer is applied to F^l , which is then concatenated with F_{local} along the channel dimension. The resultant combination undergoes processing through an MLP layer to obtain $F_{global} \in \mathbb{R}^{N_q \times C}$. Finally, F_{local} and F_{global} are concatenated along the channel dimension and then processed using an MLP. Simultaneously, the input feature F_q^{l-1} is processed by an MLP layer. These two kinds of features are combined using element-wise summation. After this combination, they are processed using an MLP. The process can be represented as follows:

$$F_{global} = \gamma(F_{local} \odot \rho(F^l)) \quad (6)$$

$$F_q^l = \phi(\xi(F_q^{l-1}) + \delta(F_{local} \odot F_{global})) \quad (7)$$

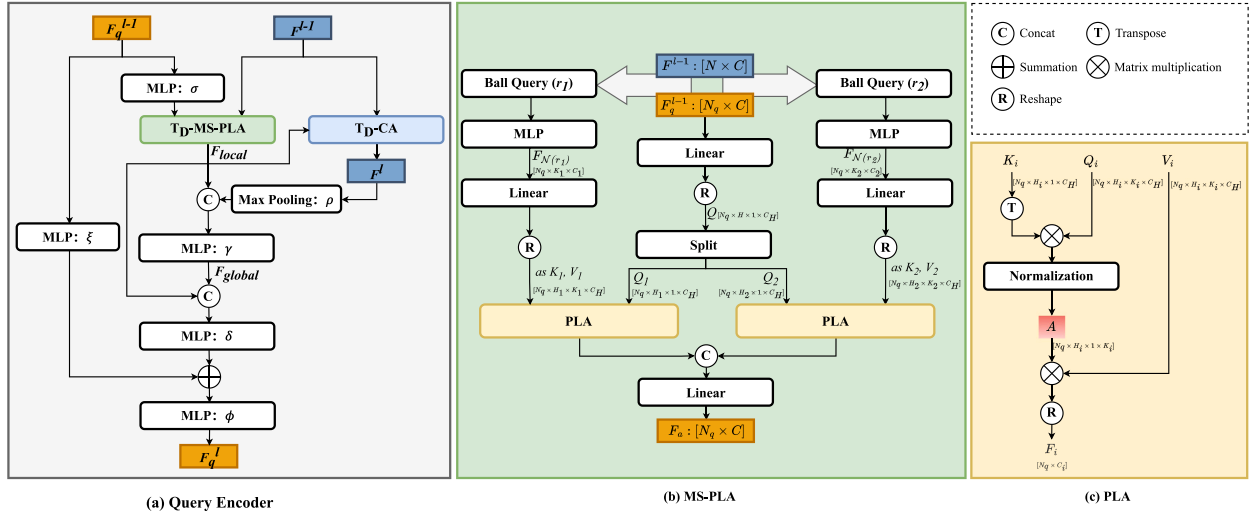


Fig. 3. Illustration of the Query Encoder. (a) (b) and (c) illustrate the structural diagram of Query Encoder, MS-PLA and PLA. Query Encoder comprises two transformer decoders embedded with MS-PLA and CA. MS-PLA extracts the local information of F_q from F , and CA uses the learned F_q to further infer F . MS-PLA employs multiple PLAs in parallel to observe local information within receptive fields at varying scales.

where, γ , ϕ , ξ and δ represents an MLP layer, ρ denotes the Max-Pooling layer, and $\odot$ means the concatenated operation.

The components denoted as T_D -MS-PLA and T_D -CA play a pivotal role in the Query Encoder. Both T_D -MS-PLA and T_D -CA adhere to a unified structure, specifically the original Transformer decoder without a self-attention layer [51]. The primary divergence lies in the configuration of the attention layer. T_D -MS-PLA incorporates the Multi-scale Point Local Attention (MS-PLA), employing multiple Point Local Attention (PLA) mechanisms in parallel. This approach facilitates the capture of local information across receptive fields of varying scales relevant to query features. In contrast, the attention layer of T_D -CA utilizes cross-attention (CA), leveraging the similarity matrix between point features and query features. This mechanism facilitates information interaction between point features and query features, consequently updating point feature information.

Multi-scale Point Local Attention. An example with two branches is illustrated in (b) of Fig. 3. For MS-PLA of the l_{th} Query Encoder, query features $F_q^{l-1} \in \mathbb{R}^{N_q \times C}$ are processed by the linear transformation and reshaping, resulting in the formation of $Q \in \mathbb{R}^{N_q \times H \times 1 \times C_H}$, where $C_H = C/H$, H denotes the number of heads. Subsequently, Q is divided into two groups according to the head dimensions of H , denoted as $Q_1 \in \mathbb{R}^{N_q \times H_1 \times 1 \times C_H}$ and $Q_2 \in \mathbb{R}^{N_q \times H_2 \times 1 \times C_H}$, where $H_1 = H_2 = H/2$. Simultaneously, we use the ball query [29] with radius of $r_i (i = 1, 2)$ to collect local features of F_q^{l-1} of F^{l-1} . We utilize these local features in a subsequent processing step involving an MLP layer, resulting in $F_{\mathcal{N}}^{l-1}(r_i) \in \mathbb{R}^{N_q \times K_i \times C_i}$. Following this, $F_{\mathcal{N}}(r_i)$ is processed by linear transformations and reshaping, yielding $K_i \in \mathbb{R}^{N_q \times H_i \times K_i \times C_H}$ and $V_i \in \mathbb{R}^{N_q \times H_i \times K_i \times C_H}$, where $C_i = H_i \times C_H$.

Then, we use PLA to conclude a further inference for semantic information. As illustrated in (c) of Fig. 3, we compute the products of Q_i with K_i^T , and apply normalization layer and softmax function to obtain the attention matrix $A \in \mathbb{R}^{N_q \times H_i \times 1 \times K_i}$. Then, the attention matrix A is multiplied by V_i to obtain the output $F_i \in \mathbb{R}^{N_q \times C_i}$ of PLA. The process of PLA can be expressed as Eq. (8). PLA utilizes neighboring features to learn semantic information. It aims to infer local correlations, so it only calculates the attention matrix between features and their neighboring features. This enables the adaptive measurement of the weight of local information, facilitating the inference of local information.

$$A = \text{softmax}\left(\frac{Q_i \otimes K_i^T}{\sqrt{C_H}}\right) \quad (8)$$

$$F_i = A \otimes V_i$$

where $\otimes$ represents the matrix multiplication, C_H represents the head channel, and Q_i , K_i and V_i are inputs of i_{th} PLA.

Subsequently, the outputs of all PLAs are concatenated. This concatenated result is then processed through a fully connected layer, resulting in the output of the MS-PLA. Therefore, by considering structural relationships at different scales, the MS-PLA effectively integrates multi-scale local information in an adaptive manner.

Cross Attention. Within the attention layer of T_D -CA, the cross attention (CA) mechanism is deployed to compute feature similarity and enact the attention mechanism. This strategic use of CA facilitates the effective establishment of relationships between the point features F and the query features F_q . By leveraging the similarity matrix derived from point features and query features, CA enables substantial information interaction between the two, consequently updating the information embedded in the point features. This iterative process involves quantifying the degree of correlation between point features and query features, and subsequently utilizing these correlation values to dynamically adjust weights. The intent is to empower the query features to guide and exert influence over the evolution of point features.

The Query Encoder leverages MS-PLA to learn features at multiple scales. This means it can capture semantic information at different scales. Through the integration of MS-PLA, the Query Encoder gains the ability to adaptively concentrate on different scales, facilitating the identification and representation of features with varying levels of detail. Additionally, the Query Encoder enables CA to capture global correlations between point features and query features. In this way, point features and query features can guide and learn from each other. To prove the effectiveness of the components of the Query Encoder, we perform ablation experiments in Section 5.4. In addition, we also conduct research on MS-PLA to verify the impact of multi-scale features on point cloud understanding tasks in Section 5.5.

3.2. Task heads

To meet the requirements of 3D point cloud understanding tasks, we specifically designed various task heads to receive the output of the Query Backbone, $F_q^{out} \in \mathbb{R}^{N_q \times C_{out}}$ and $F^{out} \in \mathbb{R}^{N \times C_{out}}$, which are obtained by learning and integrating multi-level query features F_q^l and features $F^l (l = 1, 2, \dots, L)$. These task heads include a classification head and a segmentation head.

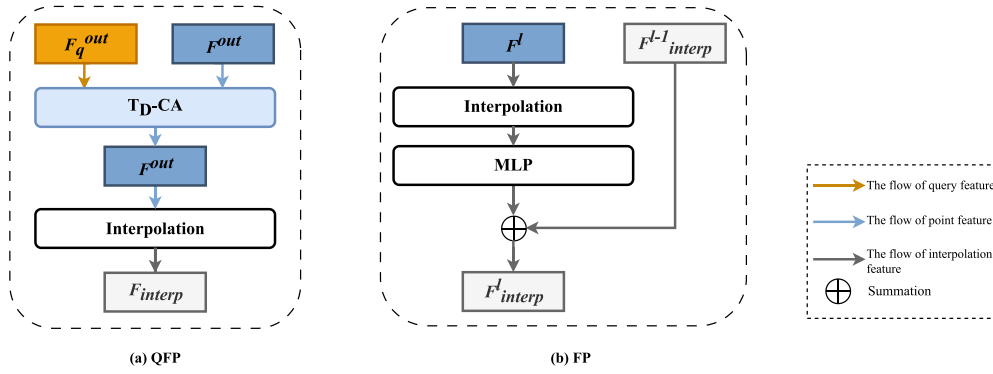


Fig. 4. Illustration of the QFP and FP. (a) denotes the QFP module, (b) means the FP module.

3.2.1. Classification head

For shape classification tasks as illustrated in the right part of Fig. 1, the classification head receives F_q^{out} and F^{out} outputted from the Query Backbone. The subsequent handling of these outputs is as follows: F_q^{out} is processed using an average pooling layer, while F^{out} passes through a max pooling layer. Following that, an element-wise addition operation amalgamates the query-specific features and global features derived from both outputs. Subsequently, integrated features are processed using a classifier. This classifier consists of two layers with LBRD (Linear-BatchNorm-ReLU-Dropout) and a linear layer to predict shape labels.

3.2.2. Segmentation head

As illustrated in the bottom part of Fig. 1, the whole process of segmentation head primarily involves two key modules: the Query Feature Propagation Module (QFP) and the Feature Propagation Module (FP). The QFP is responsible for propagating information from the query features $F_q^{out} \in \mathbb{R}^{N_q \times C_{out}}$ to the point features $F^{out} \in \mathbb{R}^{N \times C_{out}}$, and subsequently embedding this information into each point of the point cloud for $F_{interp} \in \mathbb{R}^{M \times C_{out}}$. Following this, the FP propagates the information of features $F^l \in \mathbb{R}^{N \times C}$ from the Query Encoder to update F_{interp} , obtaining segmentation labels for each point via a classifier.

Query Feature Propagation Module. A detailed structure of QFP is presented in the (a) of Fig. 4. By employing T_D -CA, F^{out} is used as the query Q , while F_q^{out} is used as both key K and value V . CA allows to learn relationship between F^{out} and F_q^{out} to infer the information in F^{out} . Then, we embed the information of F^{out} into each point of the point cloud using interpolation, obtaining $F_{interp} \in \mathbb{R}^{M \times C_{out}}$.

Feature Propagation Module. We employ the proposed FP modules, ensuring that their count matches that of Query Encoders. As illustrated in (b) of Fig. 4, for the l_{th} instance of the FP module, it receives point features $F^l \in \mathbb{R}^{N \times C}$ from the Query Encoder and the $F^{l-1}_{interp} \in \mathbb{R}^{M \times C_{out}}$ generated by the previous FP module. FP module applies the interpolation function [29] to F^l and then uses MLP to update the dimension from C to C_{out} . Subsequently, the result, referred to as $F^{l-1}_{interp} \in \mathbb{R}^{M \times C_{out}}$, is obtained by adding it to F^{l-1}_{interp} .

4. Experiments

In this section, we demonstrate the effectiveness of our network and provide quantitative and qualitative results for further analysis. We conduct shape classification, part segmentation and semantic segmentation tasks.

4.1. Shape classification on ModelNet40

Dataset. In the field of 3D computer vision, ModelNet40 [62] is often used to evaluate the performance of point cloud classification networks. The ModelNet40 dataset contains 12,311 CAD models from 40 categories. The official dataset split consists of 9843 models for

training and 2468 models for testing. Following the configuration of PointNet, 1024 points are uniformly sampled from CAD models to generate point clouds.

Quantitative and qualitative comparisons. Following the configuration of PointNet [28], each input point cloud comprises 1024 points. Instead of performing the FPS operation, we employ DGCNN [61] to extract point features $F \in \mathbb{R}^{N \times C}$, where $N = 1024$ and $C = 256$. Subsequently, utilizing the Heatmap Module, we obtain $F_q \in \mathbb{R}^{N_q \times C}$, with $N_q = 128$. Finally, we employ three Query Encoders to extract multi-level features from both F and F_q to get $F^{out} \in \mathbb{R}^{N \times C_{out}}$ and $F_q^{out} \in \mathbb{R}^{N_q \times C_{out}}$, where $C_{out} = 256 \times 3$. Each Query Encoder is composed of T_D -MS-PLA and T_D -CA. Among them, MS-PLA extracts local information at scales [0.2, 0.4], the input dimension is 256, and the number of heads is 8. The input dimension of CA is 256, and the number of heads is 8. The classifier sequentially uses 2 LBR (Linear-BN-ReLU) and an Linear layer, where the parameters of Linears are [512, 256, 40].

We evaluated our proposed network against existing methods on ModelNet40 [62], using accuracy as the evaluation metric. The classification accuracy results are presented in Table 1. HQ-Net achieves a notably high overall accuracy of 93.3% when utilizing 1k points as input, demonstrating its proficiency in point cloud classification tasks. PointNet [28], the first deep learning network for direct processing of raw point cloud data, operates on individual input points independently. PointNet++ [29] improves performance by 2.9% by introducing the sampling and grouping (SG) to extract local information. To further improve performance, KPConv [32] uses kernel functions to calculate weight matrices for points in the sphere and uses these matrices to learn point features, thereby improving classification accuracy by 0.8%. Compared to KPConv [32], our approach employs MS-PLA to observe structural information at different scales, adaptively extracting multi-scale local information, and enhancing classification accuracy by 1.2%. This highlights the importance of considering local structure at different scales to improve performance. In addition, we also conduct an experiment with single-scale MS-PLA and achieve an accuracy of 93.1%.

Although HQ-Net does not achieve the high accuracy level of some strong baseline methods, such as PointMLP [68] and PointNext [67], its performance is still satisfactory and has acceptable computational complexity. Notably, the performance of HQ-Net is comparable to some pre-trained methods, such as Point-BERT [71] and Point-MAE [72], which achieve higher overall accuracy scores. It is worth noting that HQ-Net does not rely on these pre-training strategies.

4.2. Shape classification on ScanObjectNN

Dataset. ScanObjectNN is a comprehensive benchmark dataset that comprises approximately 15,000 objects, which are categorized into 15 classes, representing 2902 unique real-world object instances. The dataset is particularly challenging for point cloud analysis methodologies due to the inherent complexities such as background clutter,

Table 1

Results of Shape classification on ModelNet40 dataset.

Method	Input	OA (%)	Params. (M)	FLOPs (G)	Throughput (ins./s)
PointNet [28]	1k points	89.2	3.5	0.5	3169
PointNet++ [29]	1k points	90.7	1.7	4.1	347
PointNet++ [29]	5k points + normal	91.1	1.7	–	–
PCNN [63]	1k points	92.3	8.2	–	–
PointCNN [22]	1k points	92.5	0.6	–	–
KPConv [32]	7k points	92.9	15.2	–	–
DGCNN [61]	1k points	92.9	1.8	2.4	518
DensePoint [64]	1k points	93.2	0.67	–	–
PosPool [65]	5k points	93.2	14.9	–	–
PointConv [45]	1k points+normal	92.5	0.6	–	–
SpecGCN [66]	2k points+normal	92.1	–	–	–
PointNeXt [67]	1k points	93.2 $\pm$ 0.1	1.4	–	–
PointGT [60]	1k points	93.2 $\pm$ 0.3	–	–	–
PointMLP [68]	1k points	93.7 $\pm$ 0.2	13.2	15.7	191
CurveNet [69]	1k points	93.8	–	–	–
RepSurf-T [70]	1k points	94.0	–	–	–
Point-BERT [71]	1k points	93.2	–	–	–
Point-MAE [72]	1k points	93.8	–	–	–
Point-M2AE [73]	1k points	94.0	–	–	–
HQ-Net(SSG)	1k points	93.1	7.9	8.0	321
HQ-Net	1k points	93.3	7.9	8.0	315

Table 2

Results of Shape classification on ScanObjectNN dataset.

Method	mAcc	OA (%)	Params. (M)	FLOPs (G)	Throughput (ins./s)
PointNet [28]	63.4	68.2	3.5	0.5	3169
PointNet++ [29]	75.4	77.9	1.7	4.1	347
PointCNN [63]	75.1	78.5	–	–	–
DGCNN [61]	73.6	78.1	1.8	2.4	518
DRNet [74]	78.0	80.3	–	–	–
GBNet [75]	77.8	80.5	8.8	9.6	193
PRANet [76]	79.1	82.1	2.3	–	–
PointMLP [68]	83.9 $\pm$ 1.5	85.4 $\pm$ 1.3	13.2	15.7	191
PointNeXt [67]	85.8 $\pm$ 0.6	87.7 $\pm$ 0.4	1.4	–	–
Point-BERT [71]	–	83.1	–	–	–
Point-MAE [72]	–	85.2	–	–	–
RepSurf-T [70]	81.2	84.1	–	–	–
HQ-Net	83.4	85.2	5.0	6.0	366

noise, and partial occlusions that are prevalent in the data. We conduct comparative experiments on the most challenging dataset PB_T50_RS.

Quantitative and qualitative comparisons. The configuration of HQ-Net, as detailed in Section 4.1, incorporates two Query Encoders to enhance its capability for dynamic feature extraction. In our comparative analysis using the challenging ScanObjectNN dataset, HQ-Net exhibits commendable performance across various contemporary point cloud classification methods. As illustrated in Table 2, HQ-Net outperforms established models such as PointNet [28] and PointNet++ [29]. This superior performance is attributed to its innovative dynamic feature extraction mechanism, which adaptively selects and processes features from the input point cloud, thereby optimizing prediction. Moreover, HQ-Net achieves comparable results to advanced models like Point-BERT [71] and Point-MAE [72], maintaining a similar OA without relying on the pre-training strategies. HQ-Net strikes an excellent balance between accuracy and computational complexity, while also demonstrating high throughput, underscoring its effectiveness in shape classification tasks. In summary, HQ-Net effectively realizes shape classification tasks in real scenes through its adaptive feature selection mechanism.

4.3. Part segmentation on ShapeNet-part

Dataset. Part segmentation is a fine-grained recognition task, which involves assigning part category labels to each point of an object. In this section, as demonstrated in Fig. 1, we employ the ShapeNet-Part dataset [21] to evaluate our part segmentation network. The ShapeNet-Part dataset [21] includes 16,881 shapes from 16 categories, with a

total of 50 parts. Most categories are labeled with 2 to 4 labeled parts. Its official split consists of 14,006 shapes for training and 2874 shapes for testing. Following the experimental configuration of PointNet [28], the part segmentation task is regarded as the classification of each point under known shape categories, and 2048 points are extracted from each shape.

Quantitative and qualitative comparisons. The configuration of the Heatmap-based Query Backbone in the part segmentation model is the same as that used in the classification task (Section 4.1). The segmentation head uses one QFP and three FP modules. QFP uses T_D -CA to learn the information of F_q^{out} to update F^{out} , where the input dimension of CA is 768 and the number of heads is 8. The parameters of MLP applied in FP are [256, 768]. The classifier sequentially uses 2 LBR (Linear-BN-ReLU) and an Linear layer, where the parameters of Linears are [512, 256, 50].

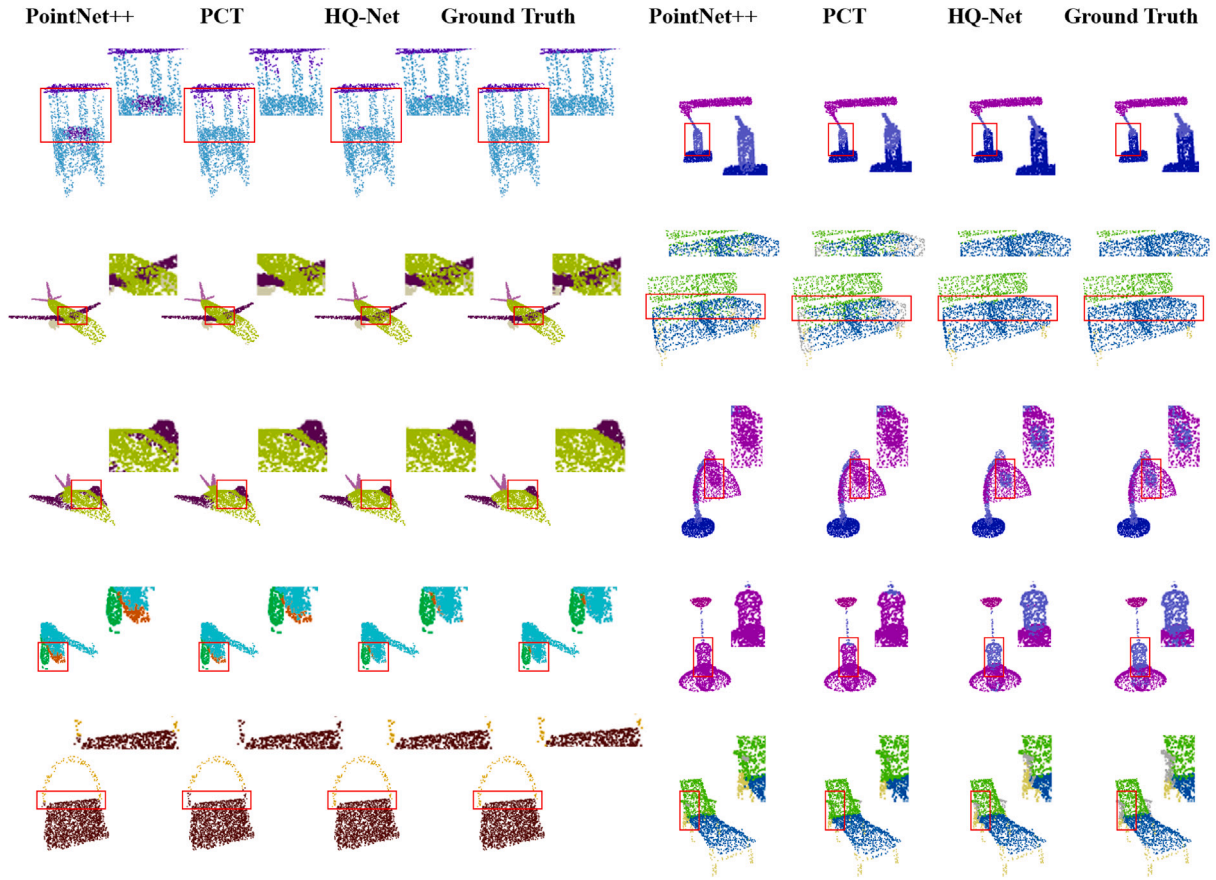
We compare our proposed network with other existing methods on the ShapeNet-Part dataset [21]. Following previous work, we use mean intersection over union (mIoU) over instances and mIoU per class as evaluation metrics. From Table 3 we can find that our network outperforms other methods on the mIoU. HQ-Net applying part segmentation achieves improvements of 1.8%, 0.5% and 0.3% over PointNet++ [29], PCT [25] and PT [11] respectively.

In Fig. 5, we present the segmentation results with regions of interest (ROI) for PointNet++ [29], PCT [25], and HQ-Net. The predictions of PointNet++ [29] and PCT [25] have wrong results, such as the chair on the left side of the first row, the pistol on the left side of the fourth row, etc. Conversely, HQ-Net achieves better prediction results. The light bulbs in the third and fourth rows of the right column are

Table 3

Results of shape segmentation on ShapeNet-Part Dataset. The evaluation metric is mIoU over instance(ins. mIoU).

Method	ins.mIoU	Airplane	Bag	Cap	Car	Chair	Earphone	Guitar	Knife	Lamp	Laptop	Motorbike	Mug	Pistol	Rocket	Skateboard	Table
PointNet [28]	83.7	83.4	78.7	82.5	74.9	89.6	73.0	91.5	85.9	80.8	95.3	65.2	93.0	81.2	57.9	72.8	80.6
PointNet++ [29]	85.1	82.4	79.0	87.7	77.3	90.8	71.8	91.0	85.9	83.7	95.3	71.6	94.1	81.3	58.7	76.4	82.6
Kd-net [77]	82.3	80.1	74.6	74.3	70.3	88.6	73.5	90.2	87.2	81.0	94.9	57.4	86.7	78.1	51.8	69.9	80.3
SSCNN [78]	84.7	81.6	81.7	81.9	75.2	90.2	74.9	93.0	86.1	84.7	95.6	66.7	92.7	81.6	60.6	82.9	82.1
SpiderCNN [79]	85.3	83.5	81.0	87.2	77.5	90.7	76.8	91.1	87.3	83.3	95.8	70.2	93.5	82.7	59.7	75.8	82.8
DGCNN [61]	85.2	84.0	83.4	86.7	77.8	90.6	74.7	91.2	87.5	82.8	95.7	66.3	94.9	81.1	63.5	74.5	82.6
PointCNN [22]	86.1	84.1	86.5	86.0	80.8	90.6	79.7	92.3	88.4	85.3	96.1	77.2	95.3	84.2	64.2	80.0	83.0
RS-CNN [80]	86.2	83.5	84.8	88.8	79.6	91.2	81.1	91.6	88.4	86.0	96.0	73.7	94.1	83.4	60.5	77.7	83.6
PT [11]	86.6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
PointASNL [81]	86.1	84.1	84.7	87.9	79.7	92.2	73.7	91.0	87.2	84.2	95.8	74.4	95.2	81.0	63.0	76.3	83.2
PCT [25]	86.4	85.0	82.4	89.0	81.2	91.9	71.5	91.3	88.1	86.3	95.8	64.6	95.8	83.6	62.2	77.6	83.7
DSPoint [82]	85.8	84.1	84.6	88.2	79.2	90.3	77.9	91.7	88.1	81.6	95.9	72.6	94.9	84.4	64.4	80.8	83.9
Point-BERT [71]	85.6	84.3	84.8	88.0	79.8	91.0	81.7	91.6	87.9	85.2	95.6	75.6	94.7	84.3	63.4	76.3	81.5
PointGT [60]	86.7	84.9	85.7	84.4	84.5	91.8	82.4	92.0	87.8	84.6	96.3	77.4	96.1	83.7	65.2	77.5	93.8
HQ-Net (FPS)	86.6	87.6	84.8	75.8	79.3	91.4	56.6	92.7	87.2	81.3	96.5	60.2	92.0	86.8	59.7	77.6	85.4
HQ-Net	86.9	88.2	86.2	77.3	79.2	91.6	63.1	92.7	88.3	83.4	96.8	61.1	92.5	85.6	58.3	75.6	85.3

**Fig. 5.** Object part segmentation results from PointNet++, PCT, and HQ-Net.

predicted correctly by HQ-Net and are closer to the ground truth. HQ-Net leverages the power of MS-PLA to capture local information and CA to extract global correlations, effectively enhancing performance of part segmentation.

4.4. Semantic segmentation on S3DIS

Dataset. For point cloud semantic segmentation tasks, we use the S3DIS dataset [83] in our experiments. S3DIS consists of 271 point clouds from six major indoor areas, such as offices and conference rooms, with different environmental scenarios, and is annotated with 13 semantic categories, including chairs, tables, floors, etc. For the

experimental configuration, we set Area 5 as the testing set and the remaining five areas as the training set. In addition, we also implemented 6-fold cross-validation, that is, each area is used as the testing set. For the evaluation metric, we choose the mean intersection over union (mIoU) metric.

Quantitative and qualitative comparisons. The hierarchical multi-level architecture is proven to be crucial when dealing with point cloud understanding tasks [84]. This architecture can extract and utilize feature information at various levels when performing per-point prediction tasks like semantic segmentation, achieving segmentation from coarse to fine, thus producing SOTA results [25,29]. Therefore, we choose PointNet++ [29] which adopts hierarchical multi-level

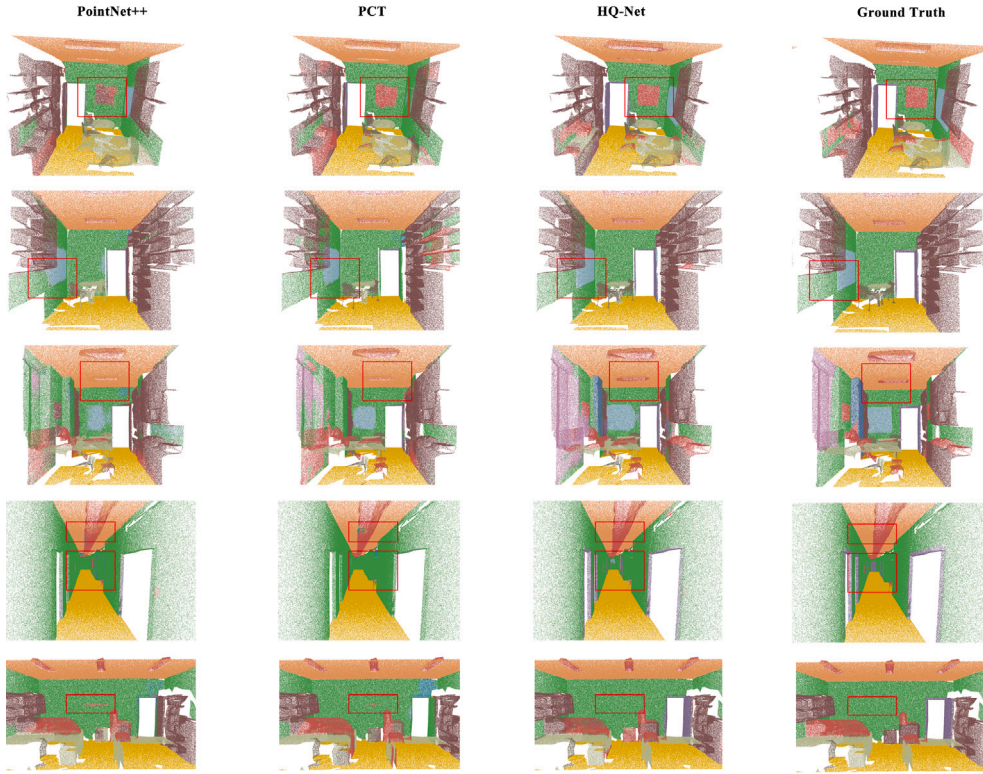


Fig. 6. Semantic segmentation results on Area 5 of S3DIS dataset from PointNet++, PCT, and HQ-Net.

Table 4

Semantic segmentation results on Area 5 of S3DIS dataset.

Method	OA (%)	ins. mIoU (%)	Ceiling	Floor	Wall	Beam	Column	Window	Door	Table	Chair	Sofa	Bookcase	Board	Clutter
PointNet [28]	–	41.1	88.8	97.3	69.8	0.1	3.9	46.3	10.8	59.0	52.6	5.9	40.3	26.4	33.2
PointNet++ [29]	83.1	54.2	90.6	96.9	75.3	0.0	6.5	57.0	19.1	75.2	69.3	61.4	51.7	59.1	42.0
PointCNN [22]	85.9	57.3	92.3	98.2	79.4	0.0	17.6	22.8	62.1	74.4	80.6	31.7	66.7	62.1	56.7
PointWeb [85]	87.0	60.3	92.0	98.5	79.4	0.0	21.1	59.7	34.8	76.3	88.3	46.9	69.3	64.9	52.5
PointANSL [81]	87.7	62.6	94.3	98.4	79.1	0.0	26.7	55.2	66.2	83.3	86.8	47.6	68.3	56.4	52.1
PCT [25]	–	61.3	92.5	98.4	80.6	0.0	19.4	61.6	48.0	85.2	76.6	67.7	46.2	67.9	52.3
SC-CNN [46]	–	63.1	93.8	98.7	80.0	0.0	17.4	55.6	50.6	76.4	88.0	66.8	71.3	64.1	56.8
DSPoint [82]	–	63.3	94.2	98.1	82.4	0.0	19.1	49.9	66.2	78.2	85.6	59.0	67.9	62.3	59.9
HQ-Net	88.9	64.2	94.8	98.1	84.4	0.8	25.9	69.3	69.0	77.7	81.7	37.2	74.5	61.8	60.2

architecture as the basic model. We replace the feature extractor of PointNet++ with the Query Encoder to learn semantic information to generate the prediction results. In our experiments, we perform qualitative and quantitative experiments on Area 5 and also implement 6-fold cross-validation. The results of semantic segmentation on Area 5 are presented in Table 4, while results of the 6-fold cross-validation on the S3DIS dataset are displayed in Table 5. For visualization, Fig. 6 provides visualizations of Area 5 from PointNet++, PCT, and HQ-Net.

As shown in Table 4, PointNet [28] exhibits suboptimal performance in specific categories, such as “Board”, owing to its independent processing of each point. Recognizing the significance of capturing local information, PointNet++ [29] introduces SG operations to enhance performance, emphasizing the importance of addressing local information capture challenges. PointCNN [22] performs remarkably well in the “Floor” and “Bookcase” categories, achieving mIoU values of 98.2% and 66.7%, respectively. In contrast, HQ-Net, leveraging Query Encoders to extract both local and global information, demonstrates improved contextual understanding. A comparative analysis with PointNet++, PCT [25], and DSPoint [82] reveals that HQ-Net achieves a higher mIoU, with increases of 10%, 2.9%, and 0.9%, respectively. Notably, HQ-Net even outperforms other models in terms of mIoU for several categories, such as “Ceiling”.

Table 5

Semantic segmentation results on S3DIS dataset with 6-fold cross-validation.

Method	OA (%)	ins. mIoU (%)
PointNet [28]	78.5	47.6
PointCNN [22]	88.1	65.4
PointWeb [85]	87.3	66.7
PointANSL [81]	88.8	68.7
PCT [25]	–	61.3
HQ-Net	89.2	70.4

As depicted in Table 5, PointNet demonstrates a commendable overall accuracy of 78.5%. However, it exhibits a relatively lower instance mean Intersection over Union (ins. mIoU) at 47.6%, suggesting potential challenges in capturing fine-grained details. PCT achieves a competitive instance mIoU (61.3%) by utilizing a transformer-based architecture to capture long-range dependencies. HQ-Net outperforms all other methods, achieving an impressive overall accuracy of 89.2% and instance mIoU of 70.4%. This superior performance can be attributed to its effective extraction of query features with the Heatmap Module and the integration of multi-scale features using transformer-based Query Encoders.

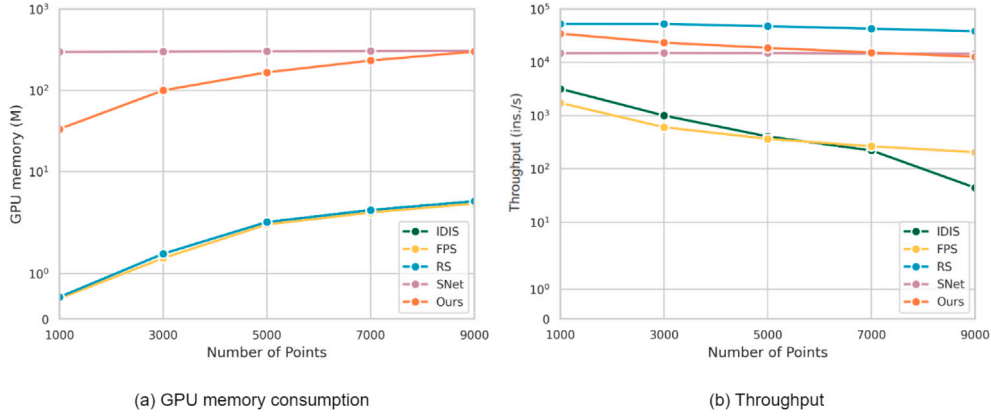


Fig. 7. Time and memory consumption of different sampling methods.

Table 6
Comparison of Sampling Methods.

Sampling Method		mAcc	OA (%)	Params. (M)	FLOPs (G)
Traditional method	FPS	90.0	92.7	4.92	5.97
	RS	89.7	92.6	4.92	5.97
	IDIS	89.6	92.3	4.92	5.97
Learning-based method	SNet	90.0	92.8	5.21	6.01
	Ours	90.3	93.1	4.96	6.01

Taking the highlighted scenario in the first line of Fig. 6 as an example, as shown in the red box, among the three networks, our network is closer to the ground truth. In the second, fourth and last rows, the highlighted parts show that the segmentation results of PointNet++ [29] and PCT [25] show some noise, while our method outperforms these two methods in terms of segmentation results. In summary, we conclude that the segmentation network composed of Query Encoders enables local and global features to guide each other during the learning process, resulting in excellent performance.

5. Ablation study

5.1. Validation of the heatmap module

In this section, we systematically evaluate various sampling methods, integrating these analyses with the performance enhancements observed when combined with our novel HQ-Net architecture. Initially, we categorize existing sampling methods into traditional and learning-based categories, such as FPS, RS, IDIS [39], and SNet [35]. We then detail how these methods are empirically compared using HQ-Net, focusing on their efficiency across different numbers of sampling points.

To assess the efficacy of these methodologies, we conduct a series of experiments utilizing the ModelNet40 dataset. Each approach is integrated into the HQ-Net framework, harnessing dual Query Encoders to facilitate meticulous inference. Evaluation ensues, employing key performance metrics such as mean accuracy (mAcc), overall accuracy (OA), model parameters, and FLOPs. As delineated in Table 6, traditional methodologies exhibit competitive performance, with accuracy spanning from 92.3% to 92.8%. This is achieved without inducing an escalation in parameter count or computational overhead. Conversely, both the learning-based SNet [35] method and our proposed approach incur an augmentation in model parameters and computational demands. In comparison with SNet [35], our HQ-Net method showcases commendable performance, boasting an mAcc of 90.3% and an OA of 93.1%. Notably, this performance enhancement is achieved with a marginal increase in parameter count by a mere 0.04, while preserving the same FLOPs.

In experiments focusing on GPU memory usage and throughput, given point clouds of different sizes, we downsample them using each sampling method, sampling approximately 25% of the points on a single GPU. Fig. 7 compares the total time and memory consumption of each sampling method for processing point clouds at different scales. When comparing the GPU memory usage and throughput of various sampling methods when processing different points, we find some key performance differences. As observed in (a), the traditional methods exhibit lower GPU memory usage, making them particularly advantageous in resource-constrained application scenarios. In comparison, the memory usage of learning-based methods such as SNet [35] is always higher, and although the memory usage of HQ-Net increases with the number of points, it is lower than that of SNet. In terms of throughput shown in (b), although the throughput of all methods decreases as the number of points increases, HQ-Net still performs better than most other methods except RS. RS exhibits the highest throughput but may not be as accurate and complex as HQ-Net in feature extraction when faced with large-scale or complex data.

5.2. Validation of the number of query features

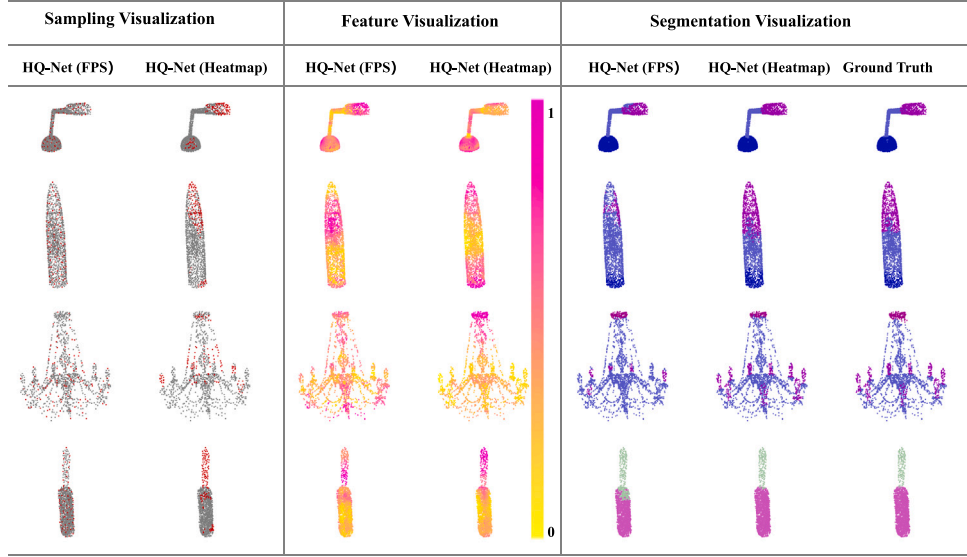
The Heatmap Module can adaptively select features based on the generated heatmap. The number of feature points affects the performance of point cloud understanding tasks. We select the efficiency of different numbers of features on the ModelNet40 dataset. In addition to our proposed method, we also additionally demonstrate the common FPS [29] algorithm. Qualitative results are then presented on the ShapeNet-Part dataset [21]. The Heatmap Module integrated into HQ-Net dynamically selects features by learning a heatmap that considers the spatial relationships and feature information of the point cloud.

According to our experimental configuration, we set the number N_q of query features (as presented in Table 7) is to 16, 48, 64, 96, 128, and 256, respectively. The Heatmap Module demonstrates significantly better performance compared to the traditional FPS method. The best performance is achieved when N_q is set to 128, resulting in a performance improvement of 0.5. It is essential to strike a balance when determining the number of query features, considering computational complexity and feature information extraction. Choosing a smaller number, like 16, may lead to inadequate feature information extraction. Conversely, opting for a larger number (e.g., 256) theoretically offer more information but can increase the computational burden. Our experimental results indicate that the model performs optimally with 128 query features.

The Fig. 8 shows the performance of HQ-Net based on Heatmap Module and HQ-Net based on FPS tested on the ShapeNet-Part dataset [21]. “Selection Visualization” in the figure represents the position of the query features on the point cloud selected using the

Table 7Validation of the number of query features on ModelNet40: The impact of different number of query features N_q .

N_q	HQ-Net (Heatmap)	Params. (M)	FLOPs (G)	Throughput (ins./s)	HQ-Net (FPS)	Params. (M)	FLOPs (G)	Throughput (ins./s)
16	93.1	7.92	5.43	408	92.6	7.87	5.39	419
48	93.1	7.92	6.17	389	92.6	7.87	6.13	368
64	93.2	7.92	6.54	366	92.7	7.87	6.50	345
96	93.2	7.92	7.28	339	92.8	7.87	7.24	310
128	93.3	7.92	8.02	315	92.8	7.87	7.99	280
256	93.2	7.92	10.99	242	93.0	7.87	10.95	202

**Fig. 8.** Visualization of results of Heatmap-based HQ-Net and FPS-based HQ-Net on ShapeNet-Part dataset.

Heatmap Module and FPS, respectively. “Feature Visualization” represents the feature visualization at each point by averaging F_{interp} . “Segmentation Visualization” represents the segmentation results of the two networks. HQ-Net based on the Heatmap Module prefers to select key features that can reflect the characteristics of the object as query features. For example, for lamps in the first and third rows, in the “Selection Visualization” part, HQ-Net focuses on the bulbs of lamps, while FPS-based HQ-Net only selects a representative subset from the point cloud. In the “Feature Visualization” section, we find that HQ-Net pays attention to each component and ensures that the feature scores of each component are basically consistent, so that the network can correctly distinguish the category of each point. In the “Segmentation Visualization”, it is proved that results of HQ-Net are closer to ground truth. The last two rows of Table 3 shows that the performance of HQ-Net based on Heatmap Module is improved by 0.3% compared to that based on FPS.

The FPS method only considers the Euclidean distance between points when selecting key points and does not combine features of the point cloud, so it has certain limitations. In contrast, Heatmap Module is a novel deep learning-based feature selection strategy that can generate heatmaps by learning point features and then dynamically select features based on this heatmap. This strategy not only considers the spatial relationship between points, but also integrates the feature information of the point cloud. Therefore, compared with the FPS method, the Heatmap Module can better capture and describe the information and structure of the point cloud.

5.3. Validation of the query encoder

Multi-level features play an important role in point cloud understanding tasks [25,29]. HQ-Net uses Query Encoder to learn multi-level features and achieve satisfactory performance. To more deeply explore the impact of multi-level features on the performance of point

cloud classification and segmentation models, we conducted a series of ablation studies.

In these experiments, we first fix all other parameters and gradually vary the number of Query Encoders. The results of Table 8 show that when L is equal to 3, the model gets the best performance. This result may indicate that the three Query Encoders are able to understand and capture the spatial structure of point cloud data. However, when L is set to 4, the model’s performance does not improve as we expected. In fact, its performance is close to that of $L=3$. This phenomenon may be due to the failure of the fourth Query Encoder to provide additional information gain, or may be due to its increased complexity leading to a certain degree of overfitting of the model.

Therefore, we can infer from the experimental results that for our task scenario, using three Query Encoders is sufficient, and increasing the number of Query Encoders cannot bring the improvement of performance. This also further emphasizes the importance of avoiding over-complication of the model, as the addition of many encoders cannot result in the expected performance gains, but can lead to an increase in model complexity and computational burden. In summary, this experiment effectively reveals the significant impact of the number of Query Encoders on model performance.

5.4. Validation of components of query encoder

The Query Encoder comprises two transformer decoders, each embedded in distinct attention layers: MS-PLA and CA. MS-PLA is focused on exploring the correlations between query features and their respective local features within specific receptive field scales. On the other hand, CA is responsible for establishing associations between the point features and query features. To assess the combined performance of these two modules, we conduct the following experiments. For the MS-PLA, we replace it with the use of multi-scale SG (MSG) in

Table 8

Validation of the number of Query Encoders on Overall Accuracy (OA) for ModelNet40 dataset and mean Intersection over Union (mIoU) on ShapeNetPart dataset. The number of parameters, FLOPs, and throughput are also reported.

<i>L</i>	ModelNet40				ShapeNet-Part			
	OA (%)	Params. (M)	FLOPs (G)	Throughput (ins./s)	ins. mIoU	Params. (M)	FLOPs (G)	Throughput (ins./s)
1	92.7	2.7	4.2	455	86.70	3.7	5.6	344
2	93.1	5.0	6.0	366	86.79	8.9	10.6	264
3	93.3	7.9	8.0	315	86.92	15.3	17.6	199
4	93.3	10.6	10.2	261	86.90	16.9	17.8	187

Table 9

Validation of components of Query Encoder on Overall Accuracy (OA). The number of parameters, FLOPs, and throughput are also reported.

Query Encoder	OA (%)	Params. (M)	FLOPs (G)	Throughput (ins./s)
MSG + MLP	92.4	4.0	8.4	301
MS-PLA + MLP	92.9	5.7	6.2	366
MSG + CA	93.1	6.1	10.2	235
MS-PLA + CA	93.3	7.9	8.0	315

Table 10

Validation of MS-PLA on Overall Accuracy (OA). The number of parameters, FLOPs, and throughput are also reported.

Level	Radius	Nsample	OA (%)	Params. (M)	FLOPs (G)	Throughput (ins./s)
1	[0.2]	[32]	93.1	7.9	8.0	340
2	[0.2, 0.4]	[32, 64]	93.3	7.9	8.0	315
3	[0.2, 0.4, 0.6]	[32, 64, 128]	93.3	9.1	10.1	234

PointNet++ [29], while keeping CA unchanged. In the case of the CA, we employ MLP in place of CA but maintained MS-PLA unchanged. Subsequently, we combined both MS-PLA and CA, which forms the Query Encoder proposed in our model.

The results presented in Table 9 indicate that the best performance is achieved when MS-PLA and CA are combined. The network composed of MS-PLA and CA outperforms the network composed of MS-PLA and MLP by 0.4%. This suggests that CA plays a crucial role in helping the model comprehend the overall structure of the point cloud data from a global perspective. The performance of the network consisting of MS-PLA and CA is 0.2% higher than the network comprised of MSG and CA. This implies that MS-PLA, leveraging the attention mechanism to extract local information at multiple scales, excels in understanding the geometric characteristics of the point cloud.

An analysis of the computational cost and processing speed reveals that MS-PLA is more efficient than MSG, while the incorporation of CA increases the computational burden. However, the combination of CA and MSG results in a significant increase in computational costs. Despite this, the pairing of MS-PLA with CA emerges as the most efficient configuration, achieving high accuracy with reasonable throughput.

In summary, these findings validate our hypothesis: MS-PLA enhances the acquisition of local information within point cloud data by establishing connections between query features and local features within specific receptive field scales. At the same time, CA, offering a global perspective, aids the model in comprehending the overall structure of the point cloud data by connecting point features and query features. Combining both MS-PLA and CA results in an improved capacity to learn and capture the intricate structure of point cloud data.

5.5. Validation of the MS-PLA

MS-PLA plays a role in the extraction of local features, employing the ball query [29] to extract local features. To assess the impact of MS-PLA on performance, we conduct experiments at various scales, as outlined in Table 10. The results clearly indicate that even at a single scale, MS-PLA outperforms some existing models [29,32]. As the scale increases, such as when the radius is set to [0.2, 0.4] and the number of local features is set to [32, 64], MS-PLA exhibits the best performance, with an improvement of 0.2%. However, with a further increase in scale, as seen in the third column, the extraction of additional information does not translate into performance gains. HQ-Net equipped

with MS-PLA can effectively learn and leverage information at different scales within point cloud data to enhance performance.

6. Conclusion

In this paper, we introduce HQ-Net, a novel point cloud understanding model aimed at improving the sampling and learning of feature points, consequently enhancing the performance of point cloud understanding tasks. The Heatmap-based Query Backbone is a key component that leverages the Heatmap Module to select point features specific to certain categories, generating heatmaps tailored to category features. This innovative approach enables automatic selection of point features based on the heatmap, leading to improved feature extraction efficiency. The integration of Query Encoders further enhances HQ-Net by facilitating mutual guidance between point features and query features. This interaction improves the model's capacity to capture both spatial structure and semantic information within the point cloud. Additionally, task-specific heads, including classification heads and segmentation heads, are incorporated to address diverse point cloud understanding tasks and provide accurate predictions. Looking ahead, our plan is to extend the applicability of HQ-Net to various 3D tasks, such as scene completion and object detection. We envision HQ-Net as a versatile pipeline that can effectively address a variety of 3D challenges.

Limitation

HQ-Net improves the sampling and learning of feature points to achieve a series of point cloud understanding tasks. However, there are the following points that can be improved:

- Heatmap Module performs well when applied to clean datasets such as ModelNet40 because it effectively highlights key features. However, its efficacy may be diminished when faced with complex and noisy real-world datasets such as ScanObjectNN. In this case, feature representation may occur, resulting in compromised model performance.
- Although the Query Encoder enhances the information exchange between the query and point features, it simultaneously increases the computational overhead. In particular, the use of T_{D-CA} mechanism increases computational requirements, posing significant challenges when processing large amounts of point cloud data.

Table A.11

GPU memory consumption for HQ-Net during training and testing.

	GPU Memory on Training (MB)	GPU Memory on Testing (MB)
Classification	224.85	133.48
Segmentation	234.12	68.73

Our focus will be on enhancing the performance of the Heatmap Module to accurately extract features in the presence of noise and incomplete data, and optimizing the T_{D-CA} mechanism to reduce computational overhead, possibly by refining the attention model or implement it using a hierarchical attention mechanism.

CRedit authorship contribution statement

Jun Li: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Shangwei Guo:** Writing – review & editing. **Luhan Wang:** Writing – review & editing. **Shaokun Han:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the China International Science and Technology Cooperation (No. 2015DFR10830).

Appendix A. GPU memory consumption for point cloud classification and segmentation

In this section, we present the GPU memory consumption for HQ-Net when performing point cloud classification and segmentation tasks. For the classification task, memory consumption is measured with a batch size of 32. For the segmentation task, the batch size is set to 16. The GPU memory usage during training and testing is summarized in Table A.11.

Appendix B. Multimodal pre-training framework with HQ-Net

Multimodal pre-training frameworks have garnered significant attention due to their exceptional performance. In this study, we investigate the effectiveness of our proposed backbone network within a multimodal pre-training framework, specifically using ULIP [86]. Due to data acquisition constraints, we focus on text and point cloud modalities (denoted as P+T). Following the ULIP framework, we evaluate zero-shot 3D classification on the complete test set of ModelNet40. The quantitative results are presented in Table B.12. HQ-Net demonstrates competitive performance with a top-1 accuracy of 46.6% and a top-5 accuracy of 68.6%, outperforming PointNet++SSG [29] and PointBERT [71] in both metrics. These results highlight the effectiveness of the proposed backbone network when integrated with the ULIP framework.

After pre-training the backbone using ULIP, we fine-tune it to perform standard 3D classification. In all experiments, we adhere to the community practice of using Overall Accuracy (OA) and Class Average Accuracy (mAcc) as evaluation metrics. The results of the standard 3D classification on ModelNet40 are presented in Table B.13. The data shows that our proposed HQ-Net backbone benefits from the ULIP

Table B.12

Results of zero-shot 3D classification on ModelNet40 for text and point cloud modalities. The results in the table are from [86].

Method	Input	top1	top5
PointNet++SSG [29] + ULIP	P+T	44.9	70.3
PointNet++MSG [29] + ULIP	P+T	48.0	63.8
PointMLP [68] + ULIP	P+T	52.2	73.0
PointBERT [71] + ULIP	P+T	44.7	66.0
HQ-Net + ULIP	P+T	46.6	68.6

Table B.13

Standard 3D classification results on ModelNet40.

Method	mAcc	OA
HQ-Net	90.7	93.3
HQ-Net + ULIP	91.0 (+0.3)	93.6 (+0.3)

Table C.14

Results on the robustness experiment of noise on ModelNet40.

Noise	0	0.005	0.01	0.02	0.03	0.04	0.05
HQ-Net	93.3	93.2	92.8	92.6	92.4	92.0	92.0

Table C.15

Results on the robustness experiment of point cloud density on ModelNet40.

npoints	400	800	1k	2k	3k	4k
HQ-Net	93.1	93.1	93.2	93.3	93.3	93.2

framework, with improved performance metrics. This improvement demonstrates that aligning representations across different modalities enhances the identification of rare categories.

In the future, we will further explore and optimize the integration of HQ-Net and the ULIP framework to tackle a series of downstream tasks such as segmentation and object detection. Additionally, we will investigate the potential benefits of multimodal pre-training for enhancing feature extraction and recognition capabilities.

Appendix C. Robustness experiments for noise and point cloud density

In this section, we evaluate the robustness of HQ-Net to noise and variations in point cloud density using the ModelNet40 dataset. To assess the impact of noise, we introduce Gaussian noise to the point clouds by scaling the point cloud coordinates with a Gaussian distribution multiplied by a constant factor. For point cloud density, we sample different numbers of points from the ModelNet40 CAD models to simulate various densities.

Table C.14 and Table C.15 present the quantitative results of the robustness tests for noise and point cloud density, respectively. Under slight noise interference, HQ-Net effectively learns the structural information of the point clouds. As the noise level increases, the geometric structure of the point clouds deteriorates, leading to a decrease in classification accuracy. Nevertheless, HQ-Net consistently maintains a high accuracy of over 92.0%. For point cloud density variations, the accuracy fluctuates between 93.0% and 93.3%, demonstrating HQ-Net's robustness to changes in point cloud density. This robustness is attributed to the Heatmap Module, which efficiently extracts category-specific features, and the Query Encoder, which enhances mutual learning between query and point features, thereby improving the model's ability to capture the spatial structure and semantic information of point clouds.

References

- [1] A. Geiger, P. Lenz, C. Stiller, R. Urtasun, Vision meets robotics: The kitti dataset, *Int. J. Robotics Res.* 32 (11) (2013) 1231–1237.

- [2] M. Jiang, Y. Wu, T. Zhao, Z. Zhao, C. Lu, Pointsift: A sift-like network module for 3d point cloud semantic segmentation, 2018, arXiv preprint arXiv:1807.00652.
- [3] Y. Park, V. Lepetit, W. Woo, Multiple 3d object tracking for augmented reality, in: 2008 7th IEEE/ACM International Symposium on Mixed and Augmented Reality, IEEE, 2008, pp. 117–120.
- [4] Y. Peng, H. Yamaguchi, Y. Funabara, S. Doki, Modeling fabric-type actuator using point clouds by deep learning, IEEE Access 10 (2022) 94363–94375.
- [5] Z. Yu, L. Li, J. Xie, C. Wang, W. Li, X. Ning, Pedestrian 3D shape understanding for person re-identification via multi-view learning, IEEE Trans. Circuits Syst. Video Technol. (2024).
- [6] C. Wang, X. Ning, W. Li, X. Bai, X. Gao, 3D person re-identification based on global semantic guidance and local feature aggregation, IEEE Trans. Circuits Syst. Video Technol. (2023).
- [7] A.H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, O. Beijbom, Pointpillars: Fast encoders for object detection from point clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 12697–12705.
- [8] Q. Hu, B. Yang, L. Xie, S. Rosa, Y. Guo, Z. Wang, N. Trigoni, A. Markham, Randla-net: Efficient semantic segmentation of large-scale point clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 11108–11117.
- [9] X. Wu, Y. Lao, L. Jiang, X. Liu, H. Zhao, Point transformer v2: Grouped vector attention and partition-based pooling, Adv. Neural Inf. Process. Syst. 35 (2022) 33330–33342.
- [10] X. Wu, L. Jiang, P.-S. Wang, Z. Liu, X. Liu, Y. Qiao, W. Ouyang, T. He, H. Zhao, Point transformer v3: Simpler, faster, stronger, 2023, arXiv preprint arXiv:2312.10035.
- [11] H. Zhao, L. Jiang, J. Jia, P.H. Torr, V. Koltun, Point transformer, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 16259–16268.
- [12] H. Xie, J. Song, Y. Zhong, C. Gu, Kalman filter finite element method for real-time soft tissue modeling, IEEE Access 8 (2020) 53471–53483.
- [13] C. Duriez, T. Bieze, Soft robot modeling, simulation and control in real-time, in: Soft Robotics: Trends, Applications and Challenges: Proceedings of the Soft Robotics Week, April 25–30, 2016, Livorno, Italy, Springer, 2017, pp. 103–109.
- [14] Y. Sato, Y. Peng, Y. Funabara, S. Doki, Funabot-finger cot: Bio-inspired worm robot for peristaltic wave locomotion and tubular structure climbing, in: 2024 IEEE/SICE International Symposium on System Integration, SII, 2024, pp. 1387–1392, <http://dx.doi.org/10.1109/SII58957.2024.10417394>.
- [15] Y. Peng, H. Nabae, Y. Funabara, K. Suzumori, Controlling a peristaltic robot inspired by inchworms, Biomim. Intell. Robotics 4 (1) (2024) 100146, <http://dx.doi.org/10.1016/j.birob.2024.100146>, URL <https://www.sciencedirect.com/science/article/pii/S2667379724000044>.
- [16] H. Su, S. Maji, E. Kalogirakis, E. Learned-Miller, Multi-view convolutional neural networks for 3d shape recognition, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 945–953.
- [17] X. Wei, R. Yu, J. Sun, View-gcn: View-based graph convolutional network for 3d shape analysis, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 1850–1859.
- [18] B. Graham, M. Engelcke, L. Van Der Maaten, 3D semantic segmentation with sub-manifold sparse convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 9224–9232.
- [19] Y. Yan, Y. Mao, B. Li, Second: Sparsely embedded convolutional detection, Sensors 18 (10) (2018) 3337.
- [20] Y. Zhou, O. Tuzel, Voxelnet: End-to-end learning for point cloud based 3d object detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 4490–4499.
- [21] Z. Wu, S. Song, A. Khosla, F. Yu, L. Zhang, J. Xiao, 3D shapenets: A deep representation for volumetric shapes, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 1912–1920.
- [22] Y. Li, R. Bu, M. Sun, W. Wu, X. Di, B. Chen, Pointcnn: Convolution on x-transformed points, Adv. Neural Inf. Process. Syst. 31 (2018).
- [23] J. Mao, X. Wang, H. Li, Interpolated convolutional networks for 3d point cloud understanding, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1578–1587.
- [24] M. Xu, R. Ding, H. Zhao, X. Qi, Paconv: Position adaptive convolution with dynamic kernel assembling on point clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 3173–3182.
- [25] M.-H. Guo, J.-X. Cai, Z.-N. Liu, T.-J. Mu, R.R. Martin, S.-M. Hu, Pct: Point cloud transformer, Comput. Vis. Media 7 (2021) 187–199.
- [26] Y. Guo, H. Wang, Q. Hu, H. Liu, L. Liu, M. Bennamoun, Deep learning for 3d point clouds: A survey, IEEE Trans. Pattern Anal. Mach. Intell. 43 (12) (2020) 4338–4364.
- [27] B. Graham, Sparse 3D convolutional neural networks, 2015, arXiv preprint arXiv:1505.02890.
- [28] C.R. Qi, H. Su, K. Mo, L.J. Guibas, Pointnet: Deep learning on point sets for 3d classification and segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 652–660.
- [29] C.R. Qi, L. Yi, H. Su, L.J. Guibas, Pointnet++: Deep hierarchical feature learning on point sets in a metric space, Adv. Neural Inf. Process. Syst. 30 (2017).
- [30] X.-F. Han, Y.-F. Jin, H.-X. Cheng, G.-Q. Xiao, Dual transformer for point cloud analysis, IEEE Trans. Multimed. (2022).
- [31] Y. Li, H. Xiong, L. Kong, R. Zhang, F. Xu, G. Chen, M. Li, MHR: MOOCs recommender service with meta hierarchical reinforced ranking, IEEE Trans. Serv. Comput. (2023).
- [32] H. Thomas, C.R. Qi, J.-E. Deschaud, B. Marcotegui, F. Goulette, L.J. Guibas, Kpconv: Flexible and deformable convolution for point clouds, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 6411–6420.
- [33] Y. Eldar, M. Lindenbaum, M. Porat, Y.Y. Zeevi, The farthest point strategy for progressive image sampling, IEEE Trans. Image Process. 6 (9) (1997) 1305–1315.
- [34] C. Moenning, N.A. Dodgson, Fast Marching Farthest Point Sampling, Tech. Rep, University of Cambridge, Computer Laboratory, 2003.
- [35] O. Dovrat, I. Lang, S. Avidan, Learning to sample, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 2760–2769.
- [36] I. Lang, A. Manor, S. Avidan, Samplenet: Differentiable point cloud sampling, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 7578–7588.
- [37] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, Q. Tian, Centernet: Keypoint triplets for object detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 6569–6578.
- [38] T. Yin, X. Zhou, P. Krahenbuhl, Center-based 3d object detection and tracking, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 11784–11793.
- [39] F. Groh, P. Wieschollek, H.P. Lensch, Flex-convolution: Million-scale point-cloud learning beyond grid-worlds, in: Asian Conference on Computer Vision, Springer, 2018, pp. 105–122.
- [40] Y. Shen, C. Feng, Y. Yang, D. Tian, Mining point cloud local structures by kernel correlation and graph pooling, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 4548–4557.
- [41] E. Nezhadarya, E. Taghavi, R. Razani, B. Liu, J. Luo, Adaptive hierarchical down-sampling for point cloud classification, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12956–12964.
- [42] S. Shi, C. Guo, L. Jiang, Z. Wang, J. Shi, X. Wang, H. Li, Pv-rcnn: Point-voxel feature set abstraction for 3d object detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 10529–10538.
- [43] K. Mo, S. Zhu, A.X. Chang, L. Yi, S. Tripathi, L.J. Guibas, H. Su, Partnet: A large-scale benchmark for fine-grained and hierarchical part-level 3d object understanding, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 909–918.
- [44] G. Riegler, A. Osman Ulusoy, A. Geiger, Octnet: Learning deep 3d representations at high resolutions, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 3577–3586.
- [45] W. Wu, Z. Qi, L. Fuxin, Pointconv: Deep convolutional networks on 3d point clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 9621–9630.
- [46] C. Wang, X. Ning, L. Sun, L. Zhang, W. Li, X. Bai, Learning discriminative features by covering local geometric space for point cloud analysis, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–15.
- [47] Y. Li, H. Xiong, Q. Wang, L. Kong, H. Liu, H. Li, J. Bian, S. Wang, G. Chen, D. Dou, et al., Coltr: Semi-supervised learning to rank with co-training and over-parameterization for web search, IEEE Trans. Knowl. Data Eng. 35 (12) (2023) 12542–12555.
- [48] Y. Li, H. Xiong, L. Kong, Z. Sun, H. Chen, S. Wang, D. Yin, Mpgraf: a modular and pre-trained graphformer for learning to rank at web-scale, in: 2023 IEEE International Conference on Data Mining, ICDM, IEEE, 2023, pp. 339–348.
- [49] Y. Li, H. Xiong, L. Kong, R. Zhang, D. Dou, G. Chen, Meta hierarchical reinforced learning to rank for recommendation: a comprehensive study in moocs, in: Joint European Conference on Machine Learning and Knowledge Discovery in Databases, Springer, 2022, pp. 302–317.
- [50] Y. Li, H. Xiong, L. Kong, J. Bian, S. Wang, G. Chen, D. Yin, GS2P: a generative pre-trained learning to rank model with over-parameterization for web-scale search, Mach. Learn. (2024) 1–19.
- [51] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, Adv. Neural Inf. Process. Syst. 30 (2017).
- [52] Y. Li, H. Xiong, L. Kong, S. Wang, Z. Sun, H. Chen, G. Chen, D. Yin, Ltrgcn: Large-scale graph convolutional networks-based learning to rank for web search, in: Joint European Conference on Machine Learning and Knowledge Discovery in Databases, Springer, 2023, pp. 635–651.
- [53] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, 2018, arXiv preprint arXiv:1810.04805.
- [54] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.

- [55] Y. Li, H. Xiong, L. Kong, Q. Wang, S. Wang, G. Chen, D. Yin, S2sphere: Semi-supervised pre-training for web search over heterogeneous learning to rank data, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023, pp. 4437–4448.
- [56] D. Lu, Q. Xie, K. Gao, L. Xu, J. Li, 3DCTN: 3D convolution-transformer network for point cloud classification, *IEEE Trans. Intell. Transp. Syst.* 23 (12) (2022) 24854–24865.
- [57] S. Qiu, S. Anwar, N. Barnes, PU-transformer: Point cloud upsampling transformer, 2021, CoRR abs/2111.12242 URL <https://arxiv.org/abs/2111.12242>.
- [58] J. Gao, J. Lan, B. Wang, F. Li, SDANet: spatial deep attention-based for point cloud classification and segmentation, *Mach. Learn.* 111 (4) (2022) 1327–1348.
- [59] X. Tang, K. Habashy, F. Huang, C. Li, D. Ban, SCA-Net: Spatial and channel attention-based network for 3D point clouds, *Comput. Vis. Image Underst.* 232 (2023) 103690.
- [60] H. Zhang, C. Wang, L. Yu, S. Tian, X. Ning, J. Rodrigues, PointGT: A method for point-cloud classification and segmentation based on local geometric transformation, *IEEE Trans. Multimed.* (2024).
- [61] Y. Wang, Y. Sun, Z. Liu, S.E. Sarma, M.M. Bronstein, J.M. Solomon, Dynamic graph cnn for learning on point clouds, *ACM Trans. Graph.* 38 (5) (2019) 1–12.
- [62] L. Yi, V.G. Kim, D. Ceylan, I.-C. Shen, M. Yan, H. Su, C. Lu, Q. Huang, A. Sheffer, L. Guibas, A scalable active framework for region annotation in 3d shape collections, *ACM Trans. Graph.* 35 (6) (2016) 1–12.
- [63] M. Atzmon, H. Maron, Y. Lipman, Point convolutional neural networks by extension operators, 2018, arXiv preprint arXiv:1803.10091.
- [64] Y. Liu, B. Fan, G. Meng, J. Lu, S. Xiang, C. Pan, Densepoint: Learning densely contextual representation for efficient point cloud processing, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 5239–5248.
- [65] Z. Liu, H. Hu, Y. Cao, Z. Zhang, X. Tong, A closer look at local aggregation operators in point cloud analysis, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXIII 16, Springer, 2020, pp. 326–342.
- [66] C. Wang, B. Samari, K. Siddiqi, Local spectral graph convolution for point set feature learning, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 52–66.
- [67] G. Qian, Y. Li, H. Peng, J. Mai, H. Hammoud, M. Elhoseiny, B. Ghanem, Pointnext: Revisiting pointnet++ with improved training and scaling strategies, *Adv. Neural Inf. Process. Syst.* 35 (2022) 23192–23204.
- [68] X. Ma, C. Qin, H. You, H. Ran, Y. Fu, Rethinking network design and local geometry in point cloud: A simple residual MLP framework, 2022, arXiv preprint arXiv:2202.07123.
- [69] A. Muzahid, W. Wan, F. Sohel, L. Wu, L. Hou, CurveNet: Curvature-based multitask learning deep networks for 3D object recognition, *IEEE/CAA J. Autom. Sin.* 8 (6) (2020) 1177–1187.
- [70] H. Ran, J. Liu, C. Wang, Surface representation for point clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 18942–18952.
- [71] X. Yu, L. Tang, Y. Rao, T. Huang, J. Zhou, J. Lu, Point-bert: Pre-training 3d point cloud transformers with masked point modeling, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 19313–19322.
- [72] Y. Pang, W. Wang, F.E. Tay, W. Liu, Y. Tian, L. Yuan, Masked autoencoders for point cloud self-supervised learning, in: European Conference on Computer Vision, Springer, 2022, pp. 604–621.
- [73] R. Zhang, Z. Guo, P. Gao, R. Fang, B. Zhao, D. Wang, Y. Qiao, H. Li, Point-m2ae: multi-scale masked autoencoders for hierarchical point cloud pre-training, *Adv. Neural Inf. Process. Syst.* 35 (2022) 27061–27074.
- [74] S. Qiu, S. Anwar, N. Barnes, Dense-resolution network for point cloud classification and segmentation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2021, pp. 3813–3822.
- [75] S. Qiu, S. Anwar, N. Barnes, Geometric back-projection network for point cloud classification, *IEEE Trans. Multimed.* 24 (2021) 1943–1955.
- [76] D.-P. Fan, G.-P. Ji, T. Zhou, G. Chen, H. Fu, J. Shen, L. Shao, Pranet: Parallel reverse attention network for polyp segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2020, pp. 263–273.
- [77] R. Klokov, V. Lempitsky, Escape from cells: Deep kd-networks for the recognition of 3d point cloud models, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 863–872.
- [78] L. Yi, H. Su, X. Guo, L.J. Guibas, Syncspecnn: Synchronized spectral cnn for 3d shape segmentation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2282–2290.
- [79] Y. Xu, T. Fan, M. Xu, L. Zeng, Y. Qiao, Spidercnn: Deep learning on point sets with parameterized convolutional filters, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 87–102.
- [80] Y. Liu, B. Fan, S. Xiang, C. Pan, Relation-shape convolutional neural network for point cloud analysis, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 8895–8904.
- [81] X. Yan, C. Zheng, Z. Li, S. Wang, S. Cui, Pointasnl: Robust point clouds processing using nonlocal neural networks with adaptive sampling, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 5589–5598.
- [82] R. Zhang, Z. Zeng, Z. Guo, X. Gao, K. Fu, J. Shi, Dspoint: Dual-scale point cloud recognition with high-frequency fusion, 2021, arXiv preprint arXiv:2111.10332.
- [83] I. Armeni, O. Sener, A.R. Zamir, H. Jiang, I. Brilakis, M. Fischer, S. Savarese, 3D semantic parsing of large-scale indoor spaces, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 1534–1543.
- [84] Z. Yang, L. Jiang, Y. Sun, B. Schiele, J. Jia, A unified query-based paradigm for point cloud understanding, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 8541–8551.
- [85] H. Zhao, L. Jiang, C.-W. Fu, J. Jia, Pointweb: Enhancing local neighborhood features for point cloud processing, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 5565–5573.
- [86] L. Xue, M. Gao, C. Xing, R. Martín-Martín, J. Wu, C. Xiong, R. Xu, J.C. Niebles, S. Savarese, Ulip: Learning a unified representation of language, images, and point clouds for 3d understanding, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 1179–1189.



Jun Li is studying for the Ph.D. degree in school of optics and photonics in Beijing Institute of Technology, Beijing, China. Her research interests include computer vision, point cloud analyze and deep learning.



Shangwei Guo received his Ph.D. degree in school of optics and photonics in Beijing Institute of Technology, Beijing, China. His research interests include LiDAR and deep learning on point clouds.



Luhan Wang received her bachelor's degree from Anhui University in 2021. Now she received the master's degree at Beijing Institute of Technology. She studies hard in computer vision and deep learning. Her research focuses on point cloud completion, classification, segmentation and so on.



Shaokun Han received his Ph.D. degree in robotics and automation systems from National Electrotechnical University, St. Petersburg, Russia, in 1997. Currently, he is a professor with Beijing Institute of Technology, Beijing, and a member of the China Instrument and Control Society. He is the author of a book and has published more than 60 papers. His research work focuses on the areas of laser ranging, laser imaging, lidar automatic target detection and recognition, lidar signal processing, and fiber-optic gyroscopes.